Answer **all** questions in the spaces provided.

1 Fig. 1.1 is a diagram of a human arm lifting an object.

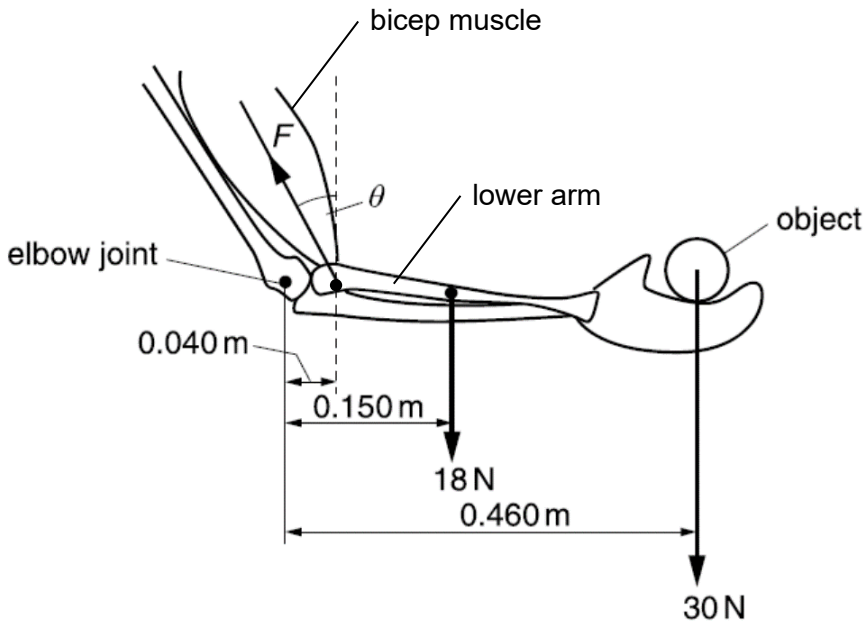

Fig. 1.1

The lower arm is horizontal and its centre of gravity is 0.150 m from the elbow joint. The weight of the lower arm is 18 N. The bicep muscle exerts a force *F* at an angle of θ to the vertical.

The horizontal distance between the elbow joint and the point of attachment of the muscle to the lower arm bone is 0.040 m. The weight of the object held in the hand is 30 N and its centre of gravity is 0.460 m from the elbow joint. The arm is in equilibrium.

(a) Define *centre of gravity*.

.....

..... [1]

(b) Determine the value of *F* when $\theta = 15°$.

F = ... N [2]

(c) For the lower arm to be in equilibrium, the elbow joint also needs to exert a force *R* on the lower arm bone.

(i) Draw a labelled arrow on Fig. 1.1 to represent the force *R* that the elbow exerts on the lower arm. [1]

(ii) Explain the direction of this force *R*.

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..... [2]

(d) As the lower arm is slowly moved away from the body in the horizontal direction, the angle θ increases.

(i) Sketch in Fig. 1.2 the graph of how *F* varies with θ. (You may assume that the lower arm remains horizontal and is in equilibrium at all times.)

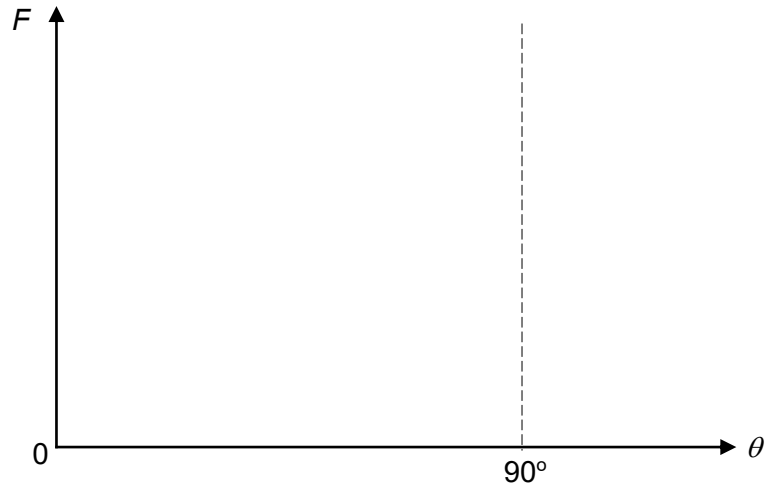

Fig. 1.2

[2]

(ii) Explain the shape of your graph in Fig. 1.2.

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..... [2]